

An information-theoretic framework for optimal experimental design in magnetic nanoparticle hyperthermia

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ABSTRACT

Magnetic nanoparticle hyperthermia is a promising cancer treatment that achieves localized tumor heating through exposure to alternating magnetic fields. However, its clinical translation is limited by uncertainties in thermophysical tissue properties, nanoparticle distribution, and intrinsic magnetic characteristics, which hinder accurate temperature prediction and treatment planning. This study presents an information-theoretic framework for optimal experimental design that systematically addresses these uncertainties to enhance the precision of hyperthermia protocols. By maximizing mutual information between measurable temperature elevations and uncertain model parameters, including tissue density, specific heat, thermal conductivity, blood perfusion rate, and nanoparticle anisotropy constant, the framework identifies optimal magnetic field amplitudes that improve parameter identifiability. A one-dimensional surrogate model of the Pennes bioheat equation, coupled with linear response theory for nanoparticle heating, enables efficient simulation of thermal outcomes across varying initial conditions and uncertainty scenarios. The method consistently converged to optimal magnetic field amplitudes across both single- and multiple-uncertain-parameter cases for a wide range of initial conditions, including constant, random, oscillatory, and near-zero magnetic fields, demonstrating robustness with respect to diverse initial conditions. Independent Monte Carlo-based optimization and verification confirmed that the proposed mutual information framework consistently identifies the same optimal mag-

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netic field protocols, demonstrating robustness with respect to the numerical integration strategy used to evaluate information gain. These findings highlight the framework's potential as a rigorous and computationally efficient tool for data-driven, uncertainty-aware experimental design in magnetic nanoparticle hyperthermia. Its application may inform improved parameter estimation and guide more effective, individualized thermal treatment strategies in both preclinical studies and future clinical translation.

1. Introduction

Cancer is a global health problem and is rising and surpassing cardiovascular and cerebrovascular diseases as the leading cause of death [1]. To address this challenge, we need better and safer treatments. Traditional treatment methods, including surgery, chemotherapy, and radiation therapy, have limitations and side effects [2]. As a result, alternative or adjuvant treatments are being investigated to improve efficacy and minimize side effects. One such approach is hyperthermia therapy, which induces a therapeutic effect by raising tissue temperature to 40–45 °C, thereby killing or eradicating cancer cells [3]. Elevated temperature activates apoptotic pathways and makes cancer cells more susceptible to chemotherapy and radiation therapy [4,5]. Recently, nanoparticle-based therapies, such as magnetic or photo-induced hyperthermia, are promising and minimally invasive options that offer targeted treatment [6]. Magnetic hyperthermia uses magnetic nanoparticles (MNPs) exposed to an alternating magnetic field (AMF) to generate localized heating within tumors, raise the temperature and damage malignant cells [7–9]. This method allows for precise targeting of tumors and less damage to surrounding healthy tissues, providing an advantage over the conventional methods [10]. A major advantage of magnetic hyperthermia is that it provides controlled and consistent heating and minimizes the risk of side effects [11]. FDA-approved superparamagnetic iron oxide nanoparticles as magnetic resonance imaging contrast agents accelerated the use of these particles in hyperthermia [12–14]. However, the clinical application of hyperthermia is hindered by several technical challenges, including uncertainties in tissue thermophysical properties, variability in intratumoral nanoparticle distribution, particle-specific magnetic parameters, and differences in heat delivery methods among various hyperthermia treatments [15]. Enhancing the precision, safety, and effectiveness of magnetic nanoparticle hyperthermia requires a systematic approach to designing treatment parameters. This process involves adjusting key variables and conditions to achieve the most efficient and effective therapeutic outcomes. Within the context of magnetic nanoparticle hyperthermia, fine-tuning nanoparticle characteristics, improving intratumoral distribution, and optimizing magnetic field parameters are crucial for maximizing therapeutic efficacy while minimizing collateral damage to healthy tissues. By employing advanced optimization techniques, greater temperature control, improved heat localization within tumors, and consistent treatment outcomes can be achieved, making hyperthermia a more reliable and effective modality for cancer therapy. Systematic refinement of parameters ensures that treatment conditions are tailored to individual patient needs, reducing variability and enhancing clinical applicability. The following sections summarize previous research efforts focused on optimizing treatment parameters to advance the clinical potential of magnetic hyperthermia and highlight further research needed to improve the efficacy of treatment.

1.1. Prior work

Hyperthermia therapy has been studied for its ability to kill cancer cells directly and boost the effectiveness of chemotherapy and radiotherapy, making it an important adjuvant therapy in multi-modality cancer treatment [16,17]. Preclinical and clinical trials in prostate cancer have shown better outcomes with hyperthermia [18,19]. These efforts highlight that it can be both standalone and adjuvant therapy. Given this potential, many studies have focused on optimizing hyperthermia treatment protocols with regard to precision, safety, and efficacy. Jiang et al. [20] optimized nanoparticle delivery by studying multi-site magnetic nanoparticle injection using the Lattice Boltzmann method combined with particle swarm optimization. They found that increasing the number of nanoparticle injection sites improved temperature homogeneity within the tumor, which is essential for uniform treatment and reducing the chance of overheating surrounding normal tissues. Wang et al. [21] developed an optimal temperature control system for tissues with gold nanoparticles using a two-energy equation model, which allows for precise temperature control by considering both conductive and convective heat transfer. Their approach minimized unintended thermal effects on surrounding tissues through periodic heating with an emphasis on controlled and localized treatment. Arora et al. [22] developed a minimum-time thermal dose control system that adjusted the treatment parameters to minimize the treatment time and prevent overheating of normal tissues. Their method was validated through simulations and in vitro experiments, showing the feasibility of time-optimal thermal therapy. They also explored the use of constrained predictive control in thermal therapies to optimize the treatment time while satisfying the safety constraints for normal tissues [23]. Barrera et al. [24] studied trapezoidal driving-field waveforms that improved specific loss power and temperature control and, hence, better cancer treatment. Abbasi and Malek [25] introduced a point-wise optimal control scheme based on the semigroup theory to refine heat delivery within tumor sites. By solving the thermal wave equation, they developed a strategy that controlled the heat deposition at the tumor location, optimizing the strength of the heat source while preventing damage to surrounding normal tissues. Although the above studies have advanced hyperthermia treatment by optimizing nanoparticle injection, temperature control, treatment efficacy, and safety for hyperthermia-based cancer therapy, the field has not systematically investigated patient-specific variability and uncertainties in thermophysical properties and nanoparticle distribution.

1.2. Gaps and motivation for this work

The effectiveness of magnetic nanoparticle hyperthermia is affected by various thermophysical properties, such as tissue density, specific heat, thermal conductivity, blood perfusion rate, and heat dissipation by particles. The efficiency of heat generation is also greatly affected by nanoparticle characteristics such as size, magnetic anisotropy constant, and magnetization strength. Conventional optimization methods in hyperthermia focus on enhancing thermal dose, typically measured through cumulative equivalent minutes at 43 °C (CEM43) or thermal damage models, while assuming constant nanoparticle properties and tissue characteristics. These assumptions do not consider patient-specific variations, which may result in suboptimal or inconsistent treatment outcomes. We propose a model-based, information-theoretic approach to optimize experimental conditions, considering the uncertainty associated with the thermophysical properties of tissues and nanoparticle parameters during hyperthermia treatment. This approach accounts for both interpatient variability and inpatient heterogeneity across different organs, thereby improving the precision and reliability of hyperthermia therapy. Additionally, for newly synthesized nanoparticles designed for hyperthermia applications, important magnetic parameters are often unknown. In these situations, it is necessary to gather experimental data in order to fit suitable models and estimate these parameters. Randomized experimental designs could result in data that provide suboptimal information on the variables of interest, increasing the uncertainty in the recovered parameters and reducing the accuracy of parameter estimation. To address these challenges, we emphasize the importance of systematically designing and optimizing experiments to maximize the information obtained from temperature measurements. This study aims to identify an optimal magnetic field amplitude and its dynamic variation during treatment to improve parameter estimation accuracy, thereby improving the precision and effectiveness of hyperthermia therapy. By integrating information-theoretic optimization strategies, this framework aims to refine experimental protocols, minimize parameter uncertainty, and advance personalized magnetic hyperthermia treatment. Specifically, the parameters considered for recovery or estimation in this study include tissue density, specific heat, thermal conductivity, blood perfusion rate, and the magnetic anisotropy constant. The information-theoretic framework introduced in this study is based on optimal experimental design methodologies, which leverage the Fisher information matrix and the Cramér-Rao bound to establish a lower limit on the variance of unbiased estimators [26,27]. Indeed, the de Bruijn identity [28] offers a fundamental link between the derivatives of entropy calculations and the Fisher information matrix, reinforcing the theoretical foundation of this approach. Conventional approaches to optimizing the Fisher information matrix require prior estimates of tissue properties, such as density, specific heat, thermal conductivity, blood perfusion rate, and the particle magnetic anisotropy constant. These estimates are iteratively refined using experimental data, necessitating repeated optimization cycles. In contrast, the framework proposed in this study directly integrates parameter uncertainties by incorporating their probability distributions into mutual information (MI) calculations. This approach enables a more systematic and unified optimization of MI, eliminating the need for repeated re-optimization. By leveraging this novel framework, the study aims to enhance the accuracy of recovered parameters, ultimately improving treatment precision and efficacy in magnetic hyperthermia applications.

2. Methods

2.1. Heat transfer model

During thermal therapy-induced hyperthermia, the temperature of the tissue is elevated beyond the normal physiological temperature limit of 37 °C. Temperature distribution within the biological tissues is a complex process involving different aspects of heat transport including heat conduction within the tissues, blood perfusion and the associated convective transfer, heat generation due to tissue metabolic activity, and the local vasculature structure. Pennes [29] presented the first mathematical model for temperature distribution in biological tissues by analyzing heat transfer in a resting human forearm. Even though Pennes' bioheat equation (PBHE) was the first mathematical model (developed in 1948), it is still widely used in hyperthermia therapy modeling due to its simplicity. The PBHE describes the spatial-temporal distribution of temperature, $T(\mathbf{x}, t)$ at a given spatial location $\mathbf{x}$ and time t within the biological tissue as:

$$\begin{aligned} \rho c \frac{\partial T}{\partial t} &= \nabla \cdot (k \nabla T) + \rho_b c_b \omega (T_b - T) + Q_m + Q_{\text{mnp}}(\mathbf{x}, t), \quad \forall \mathbf{x} \in \Omega, \\ Q_{\text{mnp}}(\mathbf{x}, t) &= \alpha \times \text{SAR} \times C_{\text{mnp}}(\mathbf{x}), \\ T(\mathbf{x}, 0) &= T_0(\mathbf{x}) = T_{\text{ss}}(\mathbf{x}), \quad \mathbf{x} \in \Omega, \\ \nabla T(\mathbf{x}, t) \cdot \mathbf{n} &= 0, \quad \mathbf{x} \in \partial\Omega_{\text{sym}}, \\ -k \nabla T(\mathbf{x}, t) \cdot \mathbf{n} &= h(T - T_{\infty}), \quad \mathbf{x} \in \partial\Omega_{\text{conv}}, \end{aligned} \quad (1)$$

where Ω is the domain volume, $\partial\Omega_{\text{sym}}$ is the symmetric boundary, $\partial\Omega_{\text{conv}}$ is the convective heat transfer boundary, ρ is the tissue density, c is the specific heat of the tissue, k is the tissue thermal conductivity, ρ_b is the blood density, c_b is the specific heat of the blood, ω is the blood perfusion rate in tissue, Q_m is the metabolic heat generation rate in the tissue, Q_{mnp} is the heat generated by the magnetic particles under the action of an external magnetic field, $\mathbf{n}$ is the outward unit normal vector, h is the heat transfer coefficient, and T_{∞} is the ambient air temperature. In this study, the metabolic heat generation rate is neglected, as the heat produced by MNPs significantly exceeds metabolic heat generation. The heat source due to the presence of nanoparticles, Q_{mnp} , is correlated to the magnetic nanoparticle concentration, C_{mnp} , and the specific absorption rate (SAR) of the nanoparticles in aqueous solution, scaled by α to account for the crowded effects when particles are in tissues.

For simplicity, we use a spatially uniform distribution of magnetic nanoparticles within the tumor domain. In practice, spatially heterogeneous nanoparticle distributions can be obtained through explicit transport modeling [30], or image-informed [31] approaches. The uniform distribution assumption is adopted to simplify the forward model and to present the proposed MI-based optimal experimental design methodology. The optimization framework is formulated in terms of the forward mapping between controllable magnetic field parameters and observable temperature measurements, allowing alternative representations of nanoparticle distribution to be incorporated as needed. The initial temperature of the tissue is denoted by $T_{ss}(\mathbf{x})$, and Neumann and Robin boundary conditions are used appropriately on the domain boundaries, where T_{ss} is the steady-state temperature distribution without external magnetic heating. One of the main challenges in using the model given in Eq. (1) is that the tissue parameters are often unknown and can vary from one patient to another [32]. Therefore, it is reasonable to consider these parameters as uncertain, with some prior knowledge about their values.

2.2. Heat generation by magnetic particles

Rosensweig [33] developed the theoretical background for understanding the heating mechanisms of magnetic nanoparticles when subjected to an external AMF. His approach was based on the Debye model [34] for dielectric dispersion in polar fluids, under the assumption that magnetic nanoparticles exhibit a linear response to the applied magnetic field. Magnetic nanoparticles, under the action of an external AMF, generate heat due to Néel [35] and Brownian [36] relaxation mechanisms. When the magnetic fluid is exposed to an AMF, the nanoparticles in the fluid rotate to align the magnetic moments in the direction of the applied magnetic field. The two different ways to achieve this alignment are Brownian motion, in which the entire particle rotates in the liquid medium, and Néel relaxation, in which the magnetic moment within the particle rotates in the direction of the applied field. The Brownian (τ_B) and Néel (τ_N) relaxation times are expressed, respectively as follows:

$$\tau_B = \frac{3\mu V_H}{\kappa_B T_{ref}}, \quad (2)$$

and

$$\tau_N = \tau_0 \left(\frac{\sqrt{\pi}}{2} \right) \left(\sqrt{\frac{\kappa_B T_{ref}}{KV_M}} \right) \exp \left(\frac{KV_M}{\kappa_B T_{ref}} \right), \quad (3)$$

where μ is magnetic fluid viscosity, V_H is the hydrodynamic volume of the nanoparticle expressed as $V_H = \frac{4\pi(a+\delta)^3}{3}$, a is the particle radius, δ is the liquid layer thickness, κ_B is Boltzmann constant, T_{ref} is nanofluid temperature which is considered to be 37 °C (normal physiological temperature), τ_0 is the attempt time which represents the characteristic time scale associated with how frequently a magnetic moment attempts to switch its orientation due to thermal fluctuations and is considered to be 10^{-9} s for iron oxide nanoparticles [37], K is the magnetic anisotropy constant, and V_M is the magnetic volume of the particle expressed as $V_M = \frac{4\pi a^3}{3}$. Since both relaxation mechanisms contribute to heat dissipation, the overall relaxation behavior of the particles is characterized by the effective relaxation time (τ), which combines both Brownian and Néel contributions as:

$$\frac{1}{\tau} = \frac{1}{\tau_B} + \frac{1}{\tau_N}. \quad (4)$$

Under the influence of an AMF, magnetic nanoparticles undergo relaxation mechanisms that result in energy dissipation as heat. This thermal energy loss, governed by Néel and Brownian relaxation processes, is quantified as volumetric power dissipation (P), which defines the rate of heat generation per unit volume when nanoparticles are dispersed in a liquid medium. For a homogeneous and monodisperse suspension of magnetic nanoparticles, the volumetric power dissipation is given by:

$$P = \mu_0 \pi f H^2 \chi'', \quad (5)$$

where μ_0 is the magnetic permeability of free space, f is the frequency of the alternating field, H is the applied magnetic field amplitude, and χ'' is the loss/imaginary component of the magnetic susceptibility, which is expressed as:

$$\chi'' = \frac{2\pi f \tau}{1 + (2\pi f \tau)^2} \chi_0, \quad (6)$$

where τ is the effective relaxation time and χ_0 is the equilibrium (static) magnetic susceptibility which quantifies how much a material becomes magnetized when exposed to an external magnetic field. The static magnetic susceptibility (χ_0) is expressed as a function of initial magnetic susceptibility (χ_i) and Langevin parameters (ξ) as follows:

$$\chi_0 = \chi_i \frac{3}{\xi} \left(\cot(\xi) - \frac{1}{\xi} \right), \quad (7)$$

$$\chi_i = \frac{\mu_0 M_d^2 \phi V_M}{3\kappa_B T_{ref}}, \quad (8)$$

$$\xi = \frac{\mu_0 M_d H V_M}{\kappa_B T_{ref}}, \quad (9)$$

where M_d is the domain magnetization of the magnetic particle. In general, the heating ability of the nanoparticles is expressed as the SAR representing the power generated per unit mass of the magnetic particle. The SAR of the nanoparticles is calculated from the volumetric heat dissipation by nanoparticles when dispersed in a liquid medium, which is expressed as:

$$\text{SAR} = \frac{P}{\rho_{\text{mnp}} \phi}, \quad (10)$$

where P is the volumetric power dissipation of the nanoparticles dispersed in the fluid, ρ_{mnp} is the mass density of the nanoparticles, and ϕ is the volume fraction of nanoparticles in the nanofluid.

2.3. Surrogate model

The first step in this investigation was to define the spatial domain and select an appropriate modeling framework. While tumor non-sphericity and geometric heterogeneity are clinically important, the present study does not aim to model patient-specific tumor morphology. Instead, a regular spherical geometry is adopted to provide a controlled computational setting for evaluating the proposed MI-based experimental design methodology. Importantly, the information-theoretic framework presented here is agnostic to model dimensionality and geometry, which enter only through the mapping between experimental design variables and observable temperature responses.

Two modeling approaches were tested for initial evaluation in this study: (i) a full three-dimensional model based on concentric spherical domains representing the tumor embedded under the skin layer, and (ii) a reduced one-dimensional axisymmetric model. The three-dimensional model was implemented and solved using both the commercial finite-element software COMSOL Multiphysics and MATLAB's PDE Toolbox. For a consistent comparison, both 3D implementations used the same spatial discretization order, time step size (20 s), and direct linear solvers, resulting in a comparable number of degrees of freedom. Under these conditions, the 3D simulation, with approximately 38,500 degrees of freedom, required 229 seconds of wall time to simulate 1800 seconds of treatment in COMSOL and 86.4 seconds in MATLAB, on a ten-core laptop with 16 GB of RAM and a 1.80 GHz processor.

Because the optimization procedure requires repeated evaluations of the forward bioheat model, using the full three-dimensional formulation would be computationally expensive. To address this limitation, a one-dimensional axisymmetric model was also simulated. The simplified model preserves the same spherical geometry and is governed by the same bioheat equation (Eq. (1)), but is solved using an in-house MATLAB implementation with a backward Euler scheme for time discretization and a central differencing scheme for spatial discretization. The resulting system contains only 56 degrees of freedom and requires approximately 0.002 s of computation time per forward solve.

The temperature distributions predicted by the one-dimensional model were compared against those obtained from both three-dimensional implementations using identical magnetic field inputs, nanoparticle properties, and thermophysical parameters at their default values. All three models produced temperature predictions in close agreement, confirming the fidelity of the one-dimensional approximation. On this basis, the one-dimensional model is referred to as a surrogate model and replaces the full three-dimensional model within the optimization procedure. The surrogate model is therefore used to enable systematic investigation of experimental design principles under repeated forward model evaluations, rather than to serve as a clinical approximation. A detailed quantitative analysis of computational efficiency and optimization wall time is presented in the Results section.

2.4. Mathematical framework for optimal experimental design

The parameters in the mathematical model, as described in Eqs. (1)-(10), that influence the temperature elevation are categorized into three groups: 1) Fixed (deterministic) parameters, which remain constant with known values, as listed in Table 1. 2) Model (uncertain) parameters, which vary between patients and nanoparticle systems. The model parameters $\mathcal{P} = \{\rho c, k, \omega, K\}$ are listed in Table 2. 3) Design parameters, defined as $\mathcal{K} = \{H_1, H_2, \dots, H_M\}$ where H_i represents the magnetic field amplitude over each time interval. As described in Eq. (5), the power deposition is directly proportional to the applied magnetic field H . During treatment, $H(t)$ is assumed to be piecewise constant over discrete time intervals Δt , leading to time-varying power dissipation. The magnetic field is therefore expressed as:

$$H(t) = \begin{cases} H_1 & t \in [0, \Delta t) \\ H_2 & t \in [\Delta t, 2\Delta t) \\ H_3 & t \in [2\Delta t, 3\Delta t) \\ \vdots & \\ H_M & t \in [(M-1)\Delta t, M\Delta t) \end{cases} \quad (11)$$

The goal of this work is to define and solve an optimization problem to determine the optimal design parameters $\mathcal{K} = \{H_1, H_2, \dots, H_M\}$, such that the uncertainty in the recovered thermophysical properties of the tumor and the magnetic anisotropy of the nanoparticle system is minimized based on available temperature measurements. To quantify the system response, we define the total signal $\mathcal{G}(\mathcal{K}, \mathcal{P}) \in \mathbb{R}$ from the physics-based model as the temperature field $T(\mathbf{x}, t)$ integrated over the spatial domain Ω and time:

$$\mathcal{G}(\mathcal{K}, \mathcal{P}) = \int_{\Omega} \int_t T(\mathbf{x}, t) dt d\Omega \quad (12)$$

Table 1

List of fixed parameters in the model.

Parameter	Value	Description
<i>Computational domain (considered in this study)</i>		
R_T	5 mm	Radius of the tumor
T_S	0.5 mm	Thickness of skin layer
<i>Blood [38]</i>		
ρ_b	1000 kg/m ³	Density
c_b	3840 J/kg·K	Specific heat
T_b	37 °C	Arterial blood temperature
<i>Magnetic particle [39]</i>		
d_{mnp}	16 nm	Particle diameter
δ	0.1 d_{mnp}	Liquid layer thickness
ρ_{mnp}	5180 kg/m ³	Particle density
M_d	446 kA/m	Magnetization
μ	1×10^{-3} Pa·s	Magnetic fluid viscosity
C_{mnp}	4 mg/cm ³ of tumor	MNP dose in tumor
V	0.1 ml	Injected magnetic nanofluid volume
<i>Constant parameters [37,40]</i>		
μ_0	$4\pi \times 10^{-7}$ H/m	Permeability of free space
κ_B	1.34×10^{-23} J/K	Boltzmann constant
τ_0	1×10^{-9} s	Attempt time
α	0.55	Correction factor
<i>Ambient parameters [30]</i>		
h	10 W/m ² ·K	Heat transfer coefficient
T_∞	24 °C	Ambient temperature
<i>Simulation parameters (considered in this study)</i>		
f	163 kHz	Field frequency
t_f	1800 s	Treatment time
M	30	Number of field on/off cycles (time steps)

The dependence of $\mathcal{G}$ on both the design parameters $\mathcal{K}$ and the model parameters $\mathcal{P}$ arises from the coupled physics described in Eqs. (1)–(10). Note that this signal incorporates the full spatiotemporal evolution of the temperature field. In the optimization framework, we treat $\mathcal{G}(\mathcal{K}, \mathcal{P})$ as the observed data. Both the data and the model parameters $\mathcal{P} \equiv \{\rho, k, \omega, K\}$ are considered as random variables. We then define the MI between the data and the model parameters and formulate an optimization problem to maximize the MI. This approach allows us to identify design parameters $\mathcal{K}$ that maximize the information gained about the uncertain model parameters from temperature measurements.

Our data acquisition model, $\mathcal{G}(\mathcal{K}, \mathcal{P}) : \mathbb{R}^a \times \mathbb{R}^m \rightarrow \mathbb{R}^n$, maps deterministic acquisition parameters, $\mathcal{K} \in \mathbb{R}^a$, and uncertain parameters, $\mathcal{P} \in \mathbb{R}^m$ to observables, $z \in \mathbb{R}^n$ (or $z \in \mathbb{C}^n$). Explicitly, we will assume that the measurement models are corrupted by zero mean white noise of a known covariance matrix, $\Sigma_z \in \mathbb{R}^{n \times n}$

$$z = \mathcal{G}(\mathcal{K}; \mathcal{P}) + v \quad v \sim \mathcal{N}(0, \Sigma_z), \quad (13)$$

where v may be interpreted as the measurement noise or the acquisition noise in the sensor model. For a deterministic measurement model $\mathcal{G}$, the conditional probability distribution has an explicit analytical form and may be written as a known Gaussian distribution.

$$p(z|\mathcal{P}) = \mathcal{N}(\mathcal{G}(\mathcal{K}; \mathcal{P}), \Sigma_z) = \frac{1}{2\pi \det \Sigma_z} \exp\left(-\frac{1}{2} \|\mathcal{G}(\mathcal{K}; \mathcal{P}) - z\|_{\Sigma_z}^2\right), \quad (14)$$

Note that the noise, Σ_z , is a property of the measurement and is fixed and independent from the signal model $\mathcal{G}(\mathcal{K}; \mathcal{P})$. Additional known information is the prior probability distributions for the model parameters, $p(\mathcal{P})$. For simplicity, we assume that prior parameters are Gaussian distributed with known mean, $\hat{\mathcal{P}}$ and covariance, Σ_p

$$\mathcal{P} \sim \mathcal{N}(\hat{\mathcal{P}}, \Sigma_p) = \frac{1}{2\pi \det \Sigma_p} \exp\left(-\frac{1}{2} \|\hat{\mathcal{P}} - \mathcal{P}\|_{\Sigma_p}^2\right), \quad (15)$$

within this Bayesian setting, Σ_p represents the model parameter variation. Bayes's theorem is fundamental to the approach. The probability of the measurements $p(z)$ must be interpreted in terms of the known information. The probability of the measurements may be derived from the marginalization of the joint probability and has the interpretation as the projection of the joint probability

Table 2Model parameters ($\mathcal{P}$) used in initialization and optimization steps.

Parameter	Default value	Range	Description
ρc	$3.5 \times 10^6 \text{ kg/m}^3$	$[2.5 \times 10^6, 4.0 \times 10^6]$	Tissue density and specific heat product
k	0.492 W/mK	$[0.300, 0.570]$	Tissue thermal conductivity
ω	0.00682 1/s	$[0.00111, 0.01390]$	Blood perfusion rate
K	31500 J/m^3	$[20000, 53000]$	Magnetic anisotropy constant

onto the measurement axis.

$$\begin{aligned}
 p(z) &= \int_{\mathcal{P}} p(\mathcal{P}, z) d\mathcal{P} = \underbrace{\int_{\mathcal{P}} p(z|\mathcal{P}) p(\mathcal{P}) d\mathcal{P}}_{\int_{\mathcal{P}} = \int_{\rho c} \int_k \int_{\omega} \int_K} \\
 &= C \int_{\mathcal{P}} d\mathcal{P} \exp\left(-\frac{1}{2}\left(\|\mathcal{G}(\mathcal{K}; \mathcal{P}) - z\|_{\Sigma_z}^2 + \|\hat{\mathcal{P}} - \mathcal{P}\|_{\Sigma_p}^2\right)\right),
 \end{aligned} \tag{16}$$

2.4.1. Mutual-information Calculation and optimization

The objective of this work is to identify experimental conditions that maximize the reliability of parameter estimates by reducing uncertainty in the recovered thermophysical and magnetic properties. Within an information-theoretic framework [28], this reduction in uncertainty is quantified by the MI, which measures the amount of information gained about one random variable through observations of another random variable. MI has been widely applied across diverse domains, including image registration [41], optimal sensor placement [42], sensor management [43], and parameter estimation in complex dynamical systems [44]. Motivated by this information-theoretic perspective, we formulate an optimal experimental design approach for magnetic nanoparticle hyperthermia, in which the applied magnetic field protocol is optimized to maximize the entropy of the temperature measurements and thereby improve parameter identifiability. Assuming Gaussian priors and likelihoods, with a measurement noise covariance Σ_z that is fixed and independent of the design variables $\mathcal{K}$, the conditional entropy $H(z | \mathcal{P})$ is constant. Consequently, maximizing the MI between the uncertain parameter vector $\mathcal{P}$ and the measurement z is equivalent to maximizing the entropy of the measurement. The experimental design problem is therefore written as,

$$\max_{\mathcal{K} \in \mathcal{F}} I(\mathcal{P}; z) = \max_{\mathcal{K} \in \mathcal{F}} H(z), \tag{17}$$

where $\mathcal{K} \in \mathcal{F}$ denotes the experimental design vector (magnetic-field protocol) constrained to the admissible design space $\mathcal{F}$.

The computation of the measurement entropy $H(z)$ is computationally demanding due to the nested high-dimensional integrals over both the parameter and data spaces. These integrals arise from marginalization of the likelihood with respect to the prior distribution. A variety of numerical strategies have been developed to approximate such integrals, including Monte Carlo and quasi-Monte Carlo methods [45,46], lattice rules [47], adaptive subdivision techniques [48], neural-network-based approximations [49], and numerical quadrature approaches [50,51]. In this work, Gauss-Hermite quadrature [51] is employed to evaluate the entropy integrals, exploiting the Gaussian structure of the prior and likelihood distributions (see Appendix A.1 for details). The entropy is approximated as

$$\begin{aligned}
 H(z) &= - \int_z p(z) \ln p(z) dz \approx - \frac{1}{N_{out}} \sum_i^{N_{out}} \ln p(z_i), \\
 &= - \int_z \int_{\mathcal{P}} \prod_i^N p(z_i | \mathcal{P}) p(\mathcal{P}) \ln \left(\int_{\hat{\mathcal{P}}} \prod_j^N p(\hat{\mathcal{P}}) p(z_j | \hat{\mathcal{P}}) d\hat{\mathcal{P}} \right) d\mathcal{P} dz, \\
 p(z_i) &= \int_{\hat{\mathcal{P}}} p(\hat{\mathcal{P}}) p(z | \hat{\mathcal{P}}) d\hat{\mathcal{P}} \approx \frac{1}{N_{in}} \sum_j^{N_{in}} p(z_{i,j} | \hat{\mathcal{P}}_j),
 \end{aligned} \tag{18}$$

Here, the samples $\mathcal{P}_i$ are drawn from the prior distribution $p(\mathcal{P})$, the corresponding measurements z_i are drawn from the conditional distribution $p(z|\mathcal{P}_i)$, and N denotes the number of quadrature points used in the integrations.

Although the experimental design problem is naturally posed as a maximization of the entropy $H(z)$, the numerical optimization is implemented in an equivalent minimization form by minimizing the negative entropy. The optimization problem is therefore written as

$$\mathcal{K}^* = \arg \min_{\mathcal{K} \in \mathcal{F}} [-H(z)]. \tag{19}$$

To improve numerical conditioning and robustness with respect to initialization, the magnetic-field amplitudes are optimized using a scaled, dimensionless parameterization. Each component of the design vector $\mathcal{K} = \{H_1, H_2, \dots, H_M\}$ is expressed as

$$H_i = H_{\min} + u_i (H_{\max} - H_{\min}), \quad i = 1, \dots, M, \tag{20}$$

where $u_i \in [0, 1]$ are scaled design variables and $H_{\min}, H_{\max}$ denote the admissible bounds on the applied magnetic field. The maximum allowable field amplitude was determined by enforcing the Hergt-Dutz safety criterion [52], which limits the amplitude-frequency

product of the applied alternating magnetic field to $Hf \leq 5 \times 10^9 \text{ A m}^{-1} \text{ s}^{-1}$, a commonly accepted practical bound in magnetic fluid hyperthermia studies. To improve numerical robustness, weak regularization terms are introduced to address reduced sensitivity at small magnetic-field amplitudes and oscillatory protocols arising from temporal discretization, leading to the final numerical objective

$$\mathbf{u}^* = \arg \min_{\mathbf{u} \in [0,1]^M} \left[-H(z(\mathcal{K}(\mathbf{u}))) + \lambda \sum_{i=1}^{M-1} (u_{i+1} - u_i)^2 + \lambda_b \sum_{i=1}^M \frac{1}{u_i + \varepsilon} \right], \quad (21)$$

with the optimal magnetic-field protocol recovered as $\mathcal{K}^* = \mathcal{K}(\mathbf{u}^*)$. The additional terms introduce, respectively, a weak smoothness regularization that discourages nonphysical oscillations in the magnetic-field amplitude and a barrier-type penalty that prevents stagnation at zero field amplitudes. The regularization parameters λ , λ_b , and $\varepsilon \ll 1$ are set to 10^{-6} , chosen sufficiently small to avoid materially altering the physically relevant optimum while improving numerical robustness.

2.4.2. Monte carlo-based mutual information optimization

Monte Carlo integration provides an independent numerical method to verify the results of experimental design optimization and those obtained using the Gauss-Hermite quadrature technique. The approach begins with the integral expression for the expectation of a function of a random variable. For a random variable X and a function $f(X)$, the expectation is given by $\mathbb{E}_X[f(X)] = \int_{\Omega_X} f(x)p_X(x)dx$ [45]. The entropy integral may thus be represented as an expectation, following this, the entropy can be written as $H(z) = -\mathbb{E}_Z[\ln(p(z))]$. This will allow the use of Monte Carlo methods to approximate the integral numerically as $H(z) \approx \frac{-1}{N_{out}} \sum_i \ln(p(z_i))$, where z_i are samples from the marginal distribution $p(z)$. However, it is not possible to directly calculate $p(z_i)$ because it is constructed as a marginal distribution, as in Eq. (16). To address this, we use a nested Monte Carlo for estimating $p(z_i)$ for given z_i . Using the Monte Carlo numerical decomposition, we can express $p(z_i)$ (i.e. the probability of some realization z_i) as the expectation integral $p(z_i) = \mathbb{E}_P[p(z_i|P)]$. We can also estimate this quantity via a Monte Carlo method via sampling from $p_0(P)$ directly. This yields $p(z_i) = \int_P p(z_i|P)p_0(P)dP \approx \frac{1}{N_{in}} \sum_j p(z_i|p_j)$ with each p_j drawn from $p_0(P)$. The probability $p(z_0|P)$ is directly calculated from Eq. (14). Given this ability to evaluate $p(z_0)$ for individual realizations z_0 , the full differential entropy expression $H(z)$ is thus the average negative logarithm of the probabilities of realizations $p(z_i)$. The individual probabilities $p(z_i)$ are generated at the highest level, using a large number of realizations z_i from the distribution $p(z)$. Each realization z_i is generated first by sampling from $z_i \sim \int_P p(z|P)p(P)dP$. For realization z_i , we then use the Monte Carlo process described above, which involves the average of $p(z_i|p_j)$ with p_j according to samples from $p_0(P)$. The full Monte Carlo integration expression for the entropy is thus

$$H(z) = - \int_Z p(z) \ln(p(z)) dz \approx \frac{-1}{N_{out}} \sum_i \ln \left(\frac{1}{N_{in}} \sum_j p(z_i|p_j) \right), \quad (22)$$

Importantly, N_{out} and N_{in} emerge as parameters required to control the convergence rate of the approximation. For known quantities, setting both to 10^3 yields an average error of approximately 5% for known mutual entropy calculations involving known $\mathcal{K}$. This is considered sufficient for benchmarking the performance of the integrations[53], and the results presented below assume these parameter settings.

This Monte Carlo integration-based entropy estimator was embedded directly into the optimization loop, replacing the Gauss-Hermite quadrature-based entropy evaluation used in Section 2.4.1. The resulting optimization problem retains the same structure as Eq. (21), with identical bounds, scaling, and regularization terms applied to the design variables $\mathcal{K}$. To assess robustness with respect to initialization and to mitigate potential nonconvexity in the objective, optimization was performed using a *MultiStart* strategy in MATLAB, with 10 distinct initial guesses, including constant, random, and oscillatory magnetic-field protocols. Importantly, aside from the numerical approximation of the MI objective, all components of the forward model, the uncertainty specification, the admissible design space, and the optimizer settings were kept identical to those used in the Gauss-Hermite-based optimization. This ensures that any differences in the resulting optimal designs can be attributed solely to the integration strategy used for evaluating MI.

2.5. MI-based parameter dependence analysis without optimization

In addition to its use as an objective function for experimental design optimization, MI, as defined in Section 2.4.1, can serve as a diagnostic tool to assess the dependence between uncertain model parameters and observable temperature measurements under a fixed experimental protocol. For a prescribed magnetic field sequence $\mathcal{K}$, the forward bioheat model produces a scalar temperature-based observable $\mathcal{G}(\mathcal{K}, P)$. To evaluate parameter dependence without performing optimization, MI is computed by allowing a single parameter $P_i \in P$ to vary according to its prior distribution while all remaining parameters are held fixed at their default values. The MI between P_i and $\mathcal{G}$ is evaluated using the formulation introduced in Eqs. (16)-(18). Repeating this procedure for each parameter yields a set of MI values that can be used to rank parameters according to their mutual dependence with the measured data. This analysis is independent of the full optimization procedure and enables identification of influential parameters and informative measurements using simplified surrogate models.

Table 3
Computational characteristics and wall times for the three forward models.

Parameter		COMSOL 3D	MATLAB 3D	MATLAB 1D
Number of elements		220,641	216,803	55
Number of nodes (DOF)		38,515	38,696	56
Time for one forward run of PBHE (s)		≈ 229	≈ 87	≈ 0.002
Time for one mutual information calculation (s)	1 uncertain variable	NA	≈ 262 ($\mathcal{O}(87 \times 3^1)$)	≈ 0.005
	4 uncertain variables	NA	≈ 7012 ($\mathcal{O}(87 \times 3^4)$)	≈ 0.06
Time for one optimization (s)	1 uncertain variable	NA	NA	≈ 14
	4 uncertain variables	NA	NA	≈ 39

NA denotes cases not evaluated in the present study.

3. Results

3.1. Model agreement and computational performance

Temperature predictions obtained from the three forward models, namely, a full three-dimensional finite-element model implemented in COMSOL Multiphysics, an independent three-dimensional model implemented using MATLAB's PDE Toolbox, and a one-dimensional axisymmetric surrogate model implemented in MATLAB, were compared under identical magnetic field inputs, nanoparticle properties, and thermophysical parameters.

Figs. 1(a) and 1(b) illustrate the geometric correspondence between the three-dimensional domain and the one-dimensional surrogate representation. Figs. 1(c) and 1(d) quantitatively compare temporal and spatial temperature predictions across all three models. At the tumor center, all models predict rapid heating from the initial baseline temperature of approximately 35 °C to a steady-state value of approximately 63 °C within the first 600 s of heating. At the outer surface ($r = 5.5$ mm), the steady-state temperature stabilizes at approximately 59 °C, while maintaining the same transient rise time. The temperature difference between the three models at any location or time point was visually negligible in the plots, indicating good agreement between the models.

Radial temperature profiles shown in Figure 1(d) further confirm spatial agreement between the models. At $t = 0$ s, all models predict a nearly uniform temperature field at 35 °C across the tumor and skin regions. At $t = 1800$ s, the temperature decreases from 63 °C at the tumor center to 59 °C at the skin surface ($r = 5.5$ mm), corresponding to a radial temperature drop of approximately 4 °C across the domain. The one-dimensional surrogate accurately reproduces both the magnitude and gradient of the spatial temperature distribution predicted by the three-dimensional models.

The computational characteristics of the three models are summarized in Table 3. The two three-dimensional implementations employed comparable spatial discretizations, with approximately 38,500 degrees of freedom, whereas the one-dimensional surrogate was simulated with 56 degrees of freedom. For a single forward solution of the Pennes bioheat equation over a 1800 s heating protocol, the three-dimensional models required approximately 229 s (COMSOL) and 86.4 s (MATLAB), while the one-dimensional surrogate required approximately 0.002 s.

MI calculations were performed using Gauss-Hermite quadrature with three quadrature points per uncertain parameter. As a result, the number of forward model evaluations required for a single MI calculation scales as 3^m , where m is the number of uncertain parameters. This scaling is reflected in the three-dimensional MATLAB implementation, for which MI evaluation times increased from approximately 261.5 s for one uncertain parameter to approximately 7012 s for four uncertain parameters. In contrast, MI evaluation times for the one-dimensional surrogate remained on the order of milliseconds, reflecting the negligible cost of individual forward solves relative to the fixed overhead associated with probability density evaluation and entropy computation. Using the surrogate model within the optimization loop, the total wall time for one optimization run was approximately 13.5 seconds for the single-parameter uncertainty case and 38.3 seconds for the four-parameter uncertainty case. These results quantify the computational effort required for MI-based optimization under the surrogate formulation. A detailed comparison of all runtimes and degrees of freedom is provided in Table 3.

3.2. Optimization

3.2.1. Gauss-Hermite quadrature-based optimization

As described in Section 2.4, multivariate Gaussian is taken as a prior for uncertain model parameters $\mathcal{P} = \{\rho c, k, \omega, K\}$. The model parameters with the known mean and variance are as follows.

$$\text{Density and Specific Heat product: } \rho c = \mathcal{N}(3589229.92, 417740.68) \text{ J/(m}^3 \cdot \text{K)}$$

$$\text{Thermal Conductivity: } k = \mathcal{N}(0.492, 0.076) \text{ W/(m} \cdot \text{K)}$$

$$\text{Blood Perfusion Rate: } \omega = \mathcal{N}(0.00682, 0.00387) \text{ s}^{-1}$$

$$\text{Magnetic anisotropy constant: } K = \mathcal{N}(31500.00, 4725.00) \text{ J/m}^3$$

All quadrature points in the third-order Gauss-Hermite quadrature approximation for numerical integration were positive with the chosen mean and variance. The other model parameters were set as specified in Table 1. The bounds for the design variable $\mathcal{K}$ were

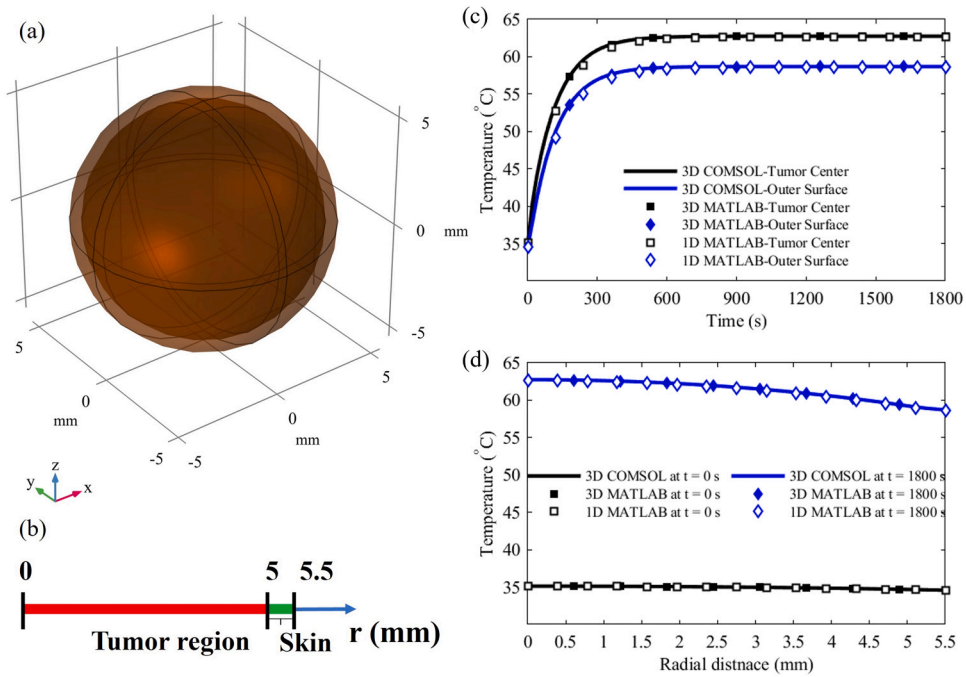


Fig. 1. Computational geometry and quantitative comparison of full and surrogate bioheat models. (a) Three-dimensional concentric spherical geometry used in the full forward models, consisting of a spherical tumor embedded within a surrounding thin skin layer. (b) One-dimensional axisymmetric surrogate representation of the same geometry, preserving spherical symmetry with a symmetry condition at tumor center ($r = 0$) and a convective boundary condition at the outer skin surface ($r = 5.5$ mm). (c) Temporal temperature evolution at the tumor center and outer tumor surface, comparing predictions from the three-dimensional COMSOL Multiphysics model, the independent three-dimensional MATLAB PDE Toolbox implementation, and the one-dimensional MATLAB surrogate model. The three models exhibit nearly identical transient heating behavior, reaching steady-state temperatures of approximately 63°C at the tumor center and 59°C at the outer surface within the first 600 s. (d) Radial temperature profiles at $t = 0$ s and $t = 1800$ s, demonstrating close agreement between all three models in both spatial gradients and absolute temperature levels. Together, these results confirm that the one-dimensional surrogate accurately reproduces the spatiotemporal thermal response of the full three-dimensional models while reducing computational cost by several orders of magnitude.

chosen to be $[0, 30,000]$ A/m. We have tried multiple initial guesses to arrive at the optimized value for $\mathcal{K}$. We have considered two cases for optimizing the design variable, as listed below.

1. *One uncertain parameter optimization.* In this case, the thermophysical properties of the tissue were assumed to be constant and fixed at the mean value. This results in the optimization problem of finding $\mathcal{K}$ that minimizes the uncertainty in the magnetic anisotropy constant of the nanoparticles.
2. *Four uncertain parameters optimization.* In this case, all four model parameters are considered uncertain, and the optimization problem is to find $\mathcal{K}$ with minimum uncertainty when these parameters are simultaneously recovered from the experimental measurements.

The optimization problem was initialized with fixed parameters, and the magnetic field amplitude H was optimized using multiple initial guesses in the interval $[0, 30,000]$ A/m. The first six initial guesses were spaced evenly, with a step size of 6000 A/m. Additionally, we explored two more cases: one with a randomly varying H and another with an oscillating H between 0 and 7600 A/m, as used in [54]. These diverse initial conditions were used to evaluate the robustness of the optimization algorithm in converging to optimal solutions. Fig. 2 and Fig. 3 illustrate the results of an optimization problem designed to determine the optimal values of $\mathcal{K}$ for cases 1 and 2, respectively. In Fig. 2, results from Case 1 are presented, where only the magnetic anisotropy constant (K) is treated as an uncertain parameter. The optimization algorithm aimed to maximize the MI between the data $\mathcal{G}(\mathcal{K}, \mathcal{P})$ and the model parameters $\mathcal{P}$. For all tested initial guesses, including zero, evenly spaced, random, and oscillatory initializations, the optimization algorithm consistently converges to the same optimal magnetic field amplitude near the maximum allowable value for the prescribed field frequency. The convergence behavior is stable and repeatable across all runs, demonstrating that the optimization is insensitive to the choice of initial guess. Fig. 3 shows the corresponding results for Case 2, in which all model parameters are treated as uncertain. Despite the increased dimensionality of the uncertainty space, the optimizer again converges to the same optimal magnetic field amplitude across all tested initial conditions. Sub-panels (a)–(f) demonstrate consistent convergence from evenly spaced initial guesses, while sub-panels (g) and (h) confirm convergence from randomly varying and oscillatory magnetic field inputs, respectively. In all cases, the optimized magnetic field amplitudes are identical and equal to the maximum allowable value.

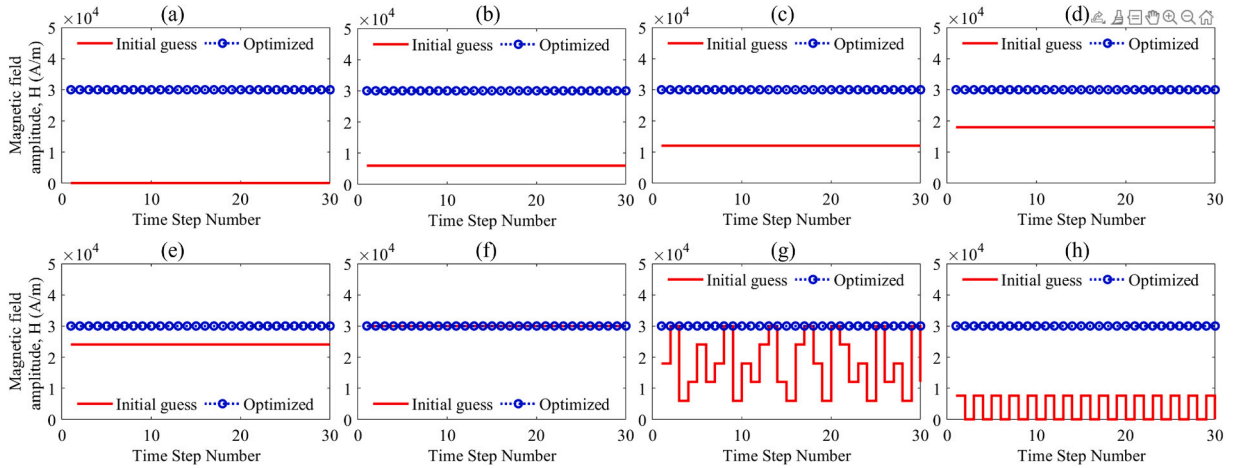


Fig. 2. Optimization results for Case 1, where only K is treated as a random variable. The plots compare the initial guesses (red solid lines) with the optimized magnetic field amplitudes (H , blue circles) across various scenarios. The initial guesses include evenly spaced values (0, 6,000, ..., 30,000 A/m) (sub-panels a-f), a randomly varying H (g), and an oscillating H between 0 and 7600 A/m (h). The optimizer consistently converges to stable and positive H values, converging to the same optimal value. When the initial guess is non-optimal, the optimizer refines it accordingly.

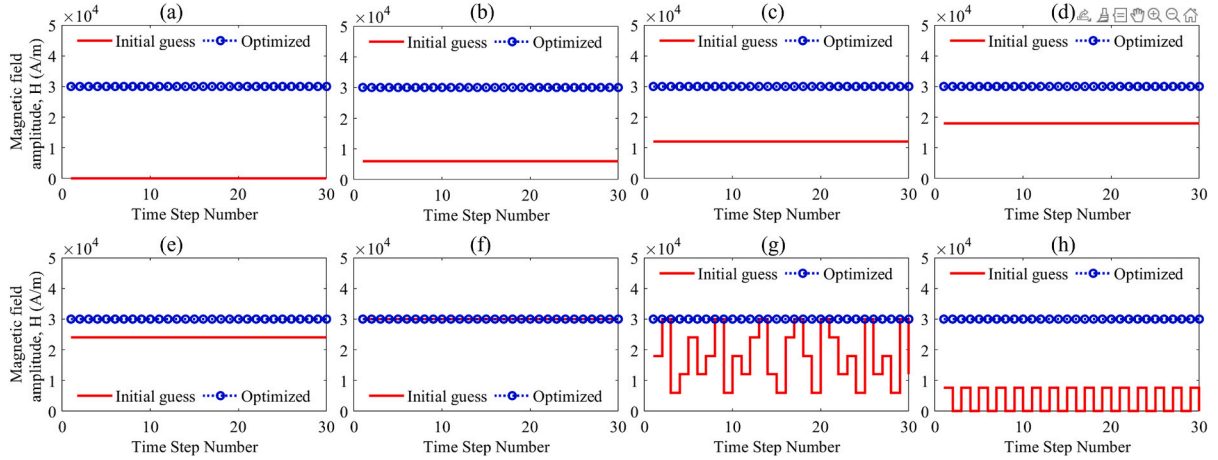


Fig. 3. Optimization results for Case 2, where all model parameters are treated as random variables. The plots compare initial guess (red solid lines) with optimized magnetic field amplitudes (H , blue circles) for evenly spaced values (0, 6,000, ..., 30,000 A/m) (sub-panels a-f), a randomly varying H (g), and an oscillating H between 0 and 7600 A/m (h). The optimizer converges to the same optimal value; when the initial guess is non-optimal, the optimizer refines it accordingly.

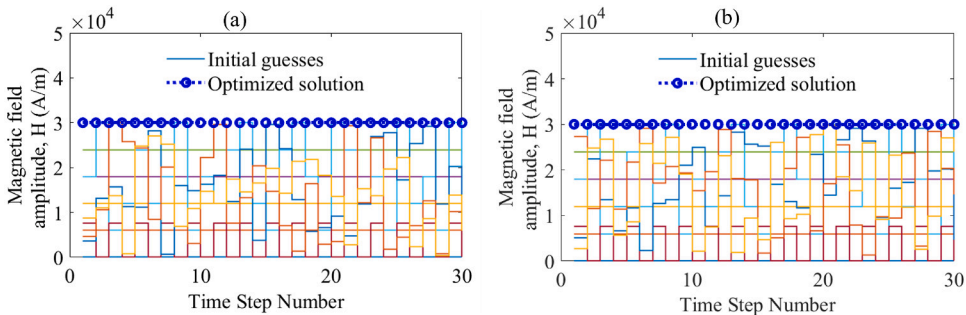


Fig. 4. Monte Carlo-based MultiStart optimization results used to verify the Gauss-Hermite quadrature-based mutual information optimization. Shown are ten distinct initial magnetic field protocols (solid lines) and the converged optimal solution (blue circles) for (a) one uncertain parameter and (b) four uncertain parameters. In all cases, the Monte Carlo integration-based optimization converges to the same optimal magnetic field amplitude identified using Gauss-Hermite quadrature, demonstrating consistency across numerical integration strategies.

3.2.2. Monte carlo-based optimization for verification

The optimal magnetic field protocols obtained using Gauss-Hermite quadrature were further examined by performing full MI-based optimization using Monte Carlo integration. Fig. 4 shows the optimal values obtained using the Monte Carlo integration-based optimization for both the single-uncertain-parameter case and the four-uncertain-parameter case, using 10 different initial guesses.

Across all initializations, including constant, random, and oscillatory magnetic field profiles, the Monte Carlo integration-based optimization consistently converged to the same optimal magnetic field amplitude identified using the Gauss-Hermite quadrature integration-based approach. This convergence behavior was observed for both uncertainty scenarios, indicating that the resulting optimal design is robust with respect to both the choice of numerical integration strategy and the initial guess supplied to the optimizer.

The optimized solutions obtained via MC integration coincide with the global optima identified using Gauss-Hermite quadrature, demonstrating that the information-theoretic objective landscape is not an artifact of the quadrature approximation. Rather, the agreement between the two independent numerical strategies provides strong evidence that the identified optimal magnetic field protocol reflects an intrinsic property of the experimental design problem under the specified measurement model and uncertainty assumptions.

Together with the Monte Carlo integration-based entropy verification results, these findings confirm that the proposed MI-based optimization framework yields consistent, numerically stable solutions, independent of the integration method used to evaluate the entropy objective.

3.2.3. Mutual information-based parameter ranking

Using the optimized magnetic field protocol and the total spatio-temporal temperature observable $\mathcal{G}$, parameter-wise MI values were computed according to Eqs. (16)–(18). The resulting MI values are

$$I(k; \mathcal{G}) = 1.4430, \quad I(\omega; \mathcal{G}) = 2.0311, \quad I(\rho c; \mathcal{G}) = 2.0311, \quad I(K; \mathcal{G}) = 1.6510. \quad (23)$$

These values quantify the relative dependence between the integrated temperature response and each uncertain model parameter under the chosen experimental protocol. Among the parameters considered, the temperature observable $\mathcal{G}$ exhibits the strongest dependence on the perfusion rate ω and the volumetric heat capacity ρc , followed by the magnetic anisotropy constant K , with the weakest dependence observed for the thermal conductivity k . Accordingly, the MI-based ranking of parameter dependence is

$$\omega \approx \rho c > K > k, \quad (24)$$

4. Discussion

The forward simulations show that, for the spherical configuration considered, the one-dimensional surrogate and three-dimensional models produce identical temperature predictions, while the associated computational cost differs by several orders of magnitude. Because MI evaluation requires repeated forward model solutions and scales with both the solve time and the number of uncertain parameters through the quadrature rule, direct application of MI to high-fidelity three-dimensional models quickly becomes computationally expensive. By contrast, the one-dimensional surrogate reduces the forward solve cost sufficiently to make repeated MI evaluations tractable, enabling systematic exploration of experimental design choices and uncertainty scenarios. The surrogate is therefore used as a controlled computational testbed to examine the behavior of the MI-based optimization framework, rather than as a clinical approximation. While extension to non-spherical and patient-specific three-dimensional geometries would require additional computational resources, the underlying MI formulation and optimization strategy remain readily adaptable.

The optimization results demonstrate that the proposed MI-based framework yields consistent and physically meaningful optimal magnetic field amplitudes across a wide range of initial conditions and uncertainty scenarios. For both the single-parameter and multi-parameter uncertainty cases, the optimizer converges to the same optimal field amplitude, indicating robustness with respect to initialization and stability of the objective landscape.

The observed convergence behavior reflects the ability of the MI criterion to guide the optimization toward experimental designs that maximize the expected information content of the temperature response with respect to uncertain model parameters. Rather than relying on heuristic tuning or exhaustive search, the MI-based formulation provides a structured and repeatable approach for selecting magnetic field parameters under uncertainty. By leveraging MI, the model identifies experimental designs that maximize information about the uncertain parameters, enabling more accurate and reliable parameter estimation from temperature measurements. The agreement between Gauss-Hermite-based and Monte Carlo-based optimizations further indicates that the identified optimal magnetic field protocol is not sensitive to the numerical strategy used to evaluate MI, but instead reflects a robust optimum of the underlying experimental design problem.

The MI-based parameter ranking provides insight into the relative informativeness of the chosen temperature observable with respect to individual model parameters, independent of the optimization process. Under the total signal measurement definition, uncertainty reduction is expected to be strongest for ω and ρc , whereas k is comparatively weakly informed by the total spatio-temporal temperature signal. This suggests that alternative measurement definitions or protocol modifications may be required to improve the identifiability of weakly coupled parameters. Such MI-based diagnostic analysis is particularly useful in deterministic bioheat modeling studies, where simplified surrogate models may be employed to assess parameter relevance and measurement sensitivity prior to full optimization or high-fidelity simulation. In this context, MI serves not only as an optimization objective but also as a practical tool for guiding manual experimental design decisions.

The application of MI theory to identify optimal experimental parameters has been validated in our previous studies [51,55,56]. For example, Jha et al. [51] demonstrated computationally how information theory-based optimization could effectively identify image acquisition parameters for recovering the pyruvate-to-lactate exchange rate. Similarly, Madankan et al. [55] validated the use of MI for selecting optimal k-space locations in MRI, using phantom acquisitions as proof of concept. Mitchell et al. [56] extended this work by optimizing MRI acquisition parameters for 3D quantification with an interleaved Look-Locker sequence (3D-QALAS), validated using both mathematical phantoms and in-vivo human brain imaging. These studies collectively highlight the robustness and versatility of information theory-based approaches in optimizing experimental conditions across a wide range of applications. Based on this established foundation, we did not explicitly validate our optimized parameters in this study with the experiments. However, we employed Monte Carlo integration techniques to verify our results, which provide strong support of identified optimal protocols.

Future work will focus on validating the proposed framework through both experimental and computational studies. Experimental validation may involve magnetic hyperthermia studies in preclinical models, where temperature measurements acquired under optimized and non-optimized magnetic field protocols can be used to assess information gain and parameter uncertainty. In parallel, computational studies using synthetic temperature data with controlled noise levels will be employed to further evaluate parameter recovery and uncertainty reduction. These efforts will support the extension of the framework toward practical applications.

A limitation of this study is the theoretical nature of the data acquisition model coming from the solution of PBHE. However, the bioheat transfer model and the magnetic particle heat generation theory have been widely used in various studies [30,39,57,58] to model magnetic nanoparticle hyperthermia. These studies strongly agreed with the experimental data when parameterized with physically realistic values. Therefore, the findings presented in this work can provide valuable insights for designing and optimizing future experiments. This work uses a Gaussian prior to handle the uncertainty in thermophysical tissue characteristics and the anisotropic energy constant of nanoparticles. The study may produce a more reliable evaluation of the stability of the findings using more uncertain parameters. Furthermore, investigating different prior distributions, such as uniform priors, could offer new perspectives. The computational architecture used in this study relies on a quadrature method to evaluate MI integrals, which have intrinsic constraints. Adding extra sources of uncertainty poses issues due to the "curse of dimensionality". To overcome this issue, alternate integration approaches, such as nested sampling methods for evaluating MI integrals, may provide more efficiency and scalability.

5. Conclusions

This study presents an information-theoretic framework for optimizing experimental conditions in magnetic nanoparticle hyperthermia, addressing key challenges associated with uncertainties in the thermophysical properties, nanoparticle distribution, and magnetic parameters. By incorporating MI as a metric during the proposed optimization approach, this study optimized the magnetic field amplitude for better temperature measurements to improve the accuracy of parameter estimation. The results demonstrate a systematic and information-theoretic experimental design framework with consistent convergence to stable and meaningful solutions, even in scenarios with multiple uncertain parameters. This framework provides a powerful tool for improving the planning of magnetic nanoparticle hyperthermia treatment and highlights the potential of its integration into clinical workflows to enhance the safety, efficacy, and predictability of hyperthermia-based cancer therapies. Future work should focus on experimental validation and extend this approach to more complex 3D geometries and real-time treatment monitoring.

CRediT authorship contribution statement

Mahesh Nandyala: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization; **S. Andrew Lanham:** Software, Validation, Visualization, Writing – original draft; **Prashant K. Jha:** Writing – review & editing, Software; **Chengyue Wu:** Writing – review & editing; **John D. Hazle:** Writing – review & editing, Supervision; **Thomas E. Yankeelov:** Writing – review & editing, Supervision; **R. Jason Stafford:** Writing – review & editing, Supervision, Conceptualization; **Ahmed A. El-Gendy:** Writing – review & editing, Conceptualization; **David Fuentes:** Writing – review & editing, Supervision, Software, Methodology, Conceptualization.

Ethics statement

This study did not involve human participants, animals, or any data requiring ethical approval.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) solely to refine the language, improve clarity, and enhance readability. No AI tool was used for research design, data analysis, result interpretation, or the generation of scientific content. After using this tool, the authors thoroughly reviewed and edited the manuscript and take full responsibility for the content of the published article.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Gauss-hermite quadrature and gaussian conditional probability

A.1. Gauss-Hermite quadrature

One-dimensional Gauss-Hermite quadrature is of the form

$$\int_{-\infty}^{\infty} e^{-x^2} f(x) dx \approx \sum_i \omega_i f(x_i), \quad (\text{A.1})$$

where x_i and ω_i denote the quadrature abscissas and weights, respectively.

Consider the marginal probability density $p(z)$,

$$p(z) = \int_{\hat{p}_1} \int_{\hat{p}_2} \cdots \int_{\hat{p}_N} \prod_{j=1}^N p(\hat{p}_j) p(z|\hat{p}) d\hat{p} = \int_{\hat{p}_1} p(\hat{p}_1) \int_{\hat{p}_2} p(\hat{p}_2) \cdots \int_{\hat{p}_N} p(\hat{p}_N) p(z|\hat{p}) d\hat{p}. \quad (\text{A.2})$$

Assume a Gaussian prior on each parameter,

$$p(P_i) = \frac{1}{\sqrt{2\pi} \sigma_{P_i}} \exp\left(-\frac{1}{2} \left(\frac{P_i - \mu_{P_i}}{\sigma_{P_i}}\right)^2\right), \quad i = 1, \dots, N. \quad (\text{A.3})$$

For each parameter component,

$$\int_{P_i} p(P_i) p(z|P) dP_i = \int_{P_i} \frac{1}{\sqrt{2\pi} \sigma_{P_i}} \exp\left(-\frac{1}{2} \left(\frac{P_i - \mu_{P_i}}{\sigma_{P_i}}\right)^2\right) p(z|P) dP_i. \quad (\text{A.4})$$

Introducing the change of variables

$$x = \frac{P_i - \mu_{P_i}}{\sqrt{2\sigma_{P_i}}}, \quad P_i = \sqrt{2\sigma_{P_i}} x + \mu_{P_i}, \quad dP_i = \sqrt{2\sigma_{P_i}} dx, \quad (\text{A.5})$$

yields

$$\int_{P_i} p(P_i) p(z|P) dP_i = \int_x \frac{1}{\sqrt{\pi}} e^{-x^2} p\left(z | P_1, \dots, \sqrt{2\sigma_{P_i}} x + \mu_{P_i}, \dots, P_N\right) dx. \quad (\text{A.6})$$

Applying Gauss-Hermite quadrature,

$$\int_{P_i} p(P_i) p(z|P) dP_i \approx \sum_j \omega_j \frac{1}{\sqrt{\pi}} p\left(z | P_1, \dots, \sqrt{2\sigma_{P_i}} x_j + \mu_{P_i}, \dots, P_N\right). \quad (\text{A.7})$$

Repeating this procedure for all parameters and proceeding by induction gives

$$p(z) \approx \sum_{j_N} \omega_{j_N} \sum_{j_{N-1}} \omega_{j_{N-1}} \cdots \sum_{j_1} \omega_{j_1} \frac{1}{\sqrt{\pi}^N} p\left(z | \sqrt{2\sigma_{P_1}} x_{j_1} + \mu_{P_1}, \dots, \sqrt{2\sigma_{P_N}} x_{j_N} + \mu_{P_N}\right). \quad (\text{A.8})$$

The computation of $p(z)$ is independent of whether the conditional probability $p(z|P)$ is Gaussian or Rician; integration is performed with respect to the Gaussian parameter prior. Integration is linear,

$$\int (af(x) + bg(x) + ch(x)) dx = a \int f(x) dx + b \int g(x) dx + c \int h(x) dx. \quad (\text{A.9})$$

A.2. Gaussian conditional probability and entropy

We now move to a Gaussian conditional probability model and derive the corresponding entropy expression. The entropy of the measurement distribution will be

$$-H(z) = \int p(z) \ln p(z) dz \quad (\text{A.10})$$

Substituting the quadrature approximation of $p(z)$ into (A.10), the entropy expression becomes

$$-H(z) = \int \sum_{i_N} \omega_{i_N} \sum_{i_2} \omega_{i_2} \sum_{i_1} \omega_{i_1} \frac{1}{\sqrt{\pi}^N} p\left(z \mid \sqrt{2}\sigma_{p_1}x_{i_1} + \mu_{p_1}, \sqrt{2}\sigma_{p_2}x_{i_2} + \mu_{p_2}, \sqrt{2}\sigma_{p_N}x_{i_N} + \mu_{p_N}\right) \ln p(z) dz. \quad (\text{A.11})$$

Considering the signal total sum as a scalar Gaussian variable,

$$p(z|P) = \frac{1}{\sqrt{2\pi}\sigma_z} \exp\left(-\frac{1}{2}\left(\frac{z - \mathcal{G}(\mathcal{K}; P_{i_1, i_2, i_N})}{\sigma_z}\right)^2\right), \quad (\text{A.12})$$

and therefore

$$\begin{aligned} -H(z) = & \int \sum_{i_N} \omega_{i_N} \sum_{i_2} \omega_{i_2} \sum_{i_1} \omega_{i_1} \frac{1}{\sqrt{\pi}^N} \frac{1}{\sqrt{2\pi}\sigma_z} \\ & \times \exp\left(-\frac{1}{2}\left(\frac{z - \mathcal{G}(\mathcal{K}; P_{i_1, i_2, i_N})}{\sigma_z}\right)^2\right) \ln p(z) dz. \end{aligned} \quad (\text{A.13})$$

Introduce the change of variables

$$y = \frac{z - \mathcal{G}(\mathcal{K}; P_{i_1, i_2, i_N})}{\sqrt{2}\sigma_z}, \quad z = \sqrt{2}\sigma_z y + \mathcal{G}(\mathcal{K}; P_{i_1, i_2, i_N}), \quad dz = \sqrt{2}\sigma_z dy. \quad (\text{A.14})$$

Applying this transformation to (A.13) yields

$$\begin{aligned} -H(z) = & \int \sum_{i_N} \omega_{i_N} \sum_{i_2} \omega_{i_2} \sum_{i_1} \omega_{i_1} \frac{1}{\sqrt{\pi}^{N+1}} \exp(-y^2) \\ & \times \ln p\left(\sqrt{2}\sigma_z y + \mathcal{G}(\mathcal{K}; P_{i_1, i_2, i_N})\right) dy. \end{aligned} \quad (\text{A.15})$$

Applying Gauss-Hermite quadrature to the integral over y ,

$$\int e^{-y^2} f(y) dy \approx \sum_k \omega_k f(y_k), \quad (\text{A.16})$$

we obtain

$$\begin{aligned} -H(z) \approx & \sum_k \omega_k \sum_{i_N} \omega_{i_N} \sum_{i_2} \omega_{i_2} \sum_{i_1} \omega_{i_1} \frac{1}{\sqrt{\pi}^{N+1}} \\ & \times \ln p\left(\sqrt{2}\sigma_z y_k + \mathcal{G}(\mathcal{K}; P_{i_1, i_2, i_N})\right). \end{aligned} \quad (\text{A.17})$$

Using the quadrature representation of $p(z)$,

$$\begin{aligned} -H(z) \approx & \sum_k \omega_k \sum_{i_N} \omega_{i_N} \sum_{i_2} \omega_{i_2} \sum_{i_1} \omega_{i_1} \frac{1}{\sqrt{\pi}^{N+1}} \\ & \times \ln \left(\sum_{j_N} \omega_{j_N} \sum_{j_2} \omega_{j_2} \sum_{j_1} \omega_{j_1} \frac{1}{\sqrt{\pi}^N} p\left(\sqrt{2}\sigma_z y_k + \mathcal{G}(\mathcal{K}; P_{i_1, i_2, i_N}) \mid P_{j_1, j_2, j_N}\right) \right). \end{aligned} \quad (\text{A.18})$$

Substituting the Gaussian conditional probability explicitly,

$$\begin{aligned} -H(z) \approx & \sum_k \omega_k \sum_{i_N} \omega_{i_N} \sum_{i_2} \omega_{i_2} \sum_{i_1} \omega_{i_1} \frac{1}{\sqrt{\pi}^{N+1}} \\ & \times \ln \left(\sum_{j_N} \omega_{j_N} \sum_{j_2} \omega_{j_2} \sum_{j_1} \omega_{j_1} \frac{1}{\sqrt{\pi}^N \sqrt{2\pi}\sigma_z} \right. \\ & \left. \times \exp \left[-\left(\frac{1}{\sqrt{2}\sigma_z} \left(\sqrt{2}\sigma_z y_k + \mathcal{G}(\mathcal{K}; P_{i_1, i_2, i_N}) - \mathcal{G}(\mathcal{K}; P_{j_1, j_2, j_N}) \right) \right)^2 \right] \right). \end{aligned} \quad (\text{A.19})$$

Vectorization

$$\begin{bmatrix} \omega_{i_1} & \omega_{i_2} & \cdots & \omega_{i_N} \end{bmatrix} \ln \begin{pmatrix} \exp(y_k + g_{i_1} - g_{j_1}) & \exp(y_k + g_{i_1} - g_{j_2}) & \cdots & \exp(y_k + g_{i_1} - g_{j_N}) \\ \exp(y_k + g_{i_2} - g_{j_1}) & \exp(y_k + g_{i_2} - g_{j_2}) & \cdots & \exp(y_k + g_{i_2} - g_{j_N}) \\ \vdots & \vdots & \ddots & \vdots \\ \exp(y_k + g_{i_N} - g_{j_1}) & \exp(y_k + g_{i_N} - g_{j_2}) & \cdots & \exp(y_k + g_{i_N} - g_{j_N}) \end{pmatrix} \begin{bmatrix} \omega_{j_1} \\ \omega_{j_2} \\ \vdots \\ \omega_{j_N} \end{bmatrix} \quad (\text{A.20})$$

Maximizing the negative is minimizing the positive

$$- \int p(z) \ln p(z) dz \quad (\text{A.21})$$

Minimize the exponent is the dominant term.

$$\exp\left(-\left(\frac{1}{\sqrt{2}\sigma_z}\left(\sqrt{2}\sigma_z y_k + \mathcal{G}(\mathcal{K}; P_{i_1, i_2, i_N}) - \mathcal{G}(\mathcal{K}; P_{j_1, j_2, j_N})\right)\right)^2\right) \quad (\text{A.22})$$

Intuition: For any given P_{i_1, i_2, i_N} and P_{j_1, j_2, j_N} we want to find $\mathcal{K}$ such that the difference $\mathcal{G}(\mathcal{K}; P_{i_1, i_2, i_N}) - \mathcal{G}(\mathcal{K}; P_{j_1, j_2, j_N})$ is maximized. This will minimize/reduce the exponent the most. I.E. we want to find $\mathcal{K}$ to maximize the spread of the anticipated measurement data. This will provide the most information gain. Consider the alternative, if the value of $\mathcal{K}$ provides similar measured values for $\mathcal{G}(\mathcal{K}; P_{i_1, i_2, i_N})$ and $\mathcal{G}(\mathcal{K}; P_{j_1, j_2, j_N})$, the measurement would be ill-posed, uninformative and in a null space in some sense. Our information theoretic calculations are trying to eliminate a null space in some where where we cannot tell what parameter the measurement came from.

A.3. Computational considerations for mutual information evaluation

While deterministic bioheat models such as the Pennes equation are commonly solved using finite-element or finite-volume methods for a fixed set of parameters, the computation of mutual information (MI) introduces a substantially higher computational burden. Unlike a single forward simulation, MI evaluation requires repeated solutions of the forward model over the parameter uncertainty space.

In the present implementation, one MI evaluation requires solving the forward bioheat model only at the Gauss-Hermite quadrature nodes associated with the parameter prior. For N_p uncertain parameters and a Gauss-Hermite quadrature order of N_{GH} points per dimension, this corresponds to

$$N_{\text{forward}} = N_{GH}^{N_p}, \quad (\text{A.23})$$

forward-model simulations per MI evaluation.

All remaining operations in the MI objective, including the construction of pairwise differences between forward-model outputs, weighted summations, and logarithmic evaluations are performed in post-processing using the precomputed forward-model results. In particular, the integration over the measurement noise variable is handled analytically and algebraically within the final discrete MI expression and therefore does not introduce additional forward-model solves.

Consequently, for high-fidelity forward models such as three-dimensional transient Pennes equation solvers, the overall computational cost of MI evaluation is dominated by the $N_{GH}^{N_p}$ forward simulations, rather than by the entropy assembly itself [Algorithm 1](#).

A.4. Practical implementation workflow

Algorithm 1 Workflow for MI evaluation and optimization of $\mathcal{K}$.

- 1: **Input:** Forward model $\mathcal{G}(\mathcal{K}; P)$; uncertain parameters $\mathcal{P}$ with Gaussian priors (μ_P, Σ_P) ; Gauss-Hermite order N_{GH} ; feasible design set $\mathcal{K}$ (or bounds/constraints on $\mathcal{K}$); optimizer settings (initial guesses, tolerances, max iterations)
- 2: **Output:** Optimal design $\mathcal{K}^*$ and $\text{MI}(\mathcal{K}^*)$
- 3: Select uncertain parameters $\mathcal{P}$ and specify their Gaussian priors.
- 4: Choose N_{GH} and generate Gauss-Hermite quadrature nodes and weights for the parameter prior.
- 5: Define the measurement functional $\mathcal{M}[\cdot]$ (e.g., temperature at specified points, or a spatio-temporal temperature metric).
- 6: Initialize the optimizer with one or more starting designs $\mathcal{K}^{(0)} \in \mathcal{K}$.
- 7: **for** each optimizer iteration $\text{iter} = 0, 1, 2, \dots$ **do**
- 8: Propose a candidate design $\mathcal{K}^{(\text{iter})} \in \mathcal{K}$ (enforce bounds/constraints on $\mathcal{K}$).
- 9: **for** each parameter quadrature node P_i **do**
- 10: Solve the forward bioheat model to obtain the temperature field $T(\mathbf{x}, \tau; \mathcal{K}^{(\text{iter})}, P_i)$, where τ denotes physical time.
- 11: Compute and store the measurement $\mathcal{G}(\mathcal{K}^{(\text{iter})}) = \mathcal{M}[T(\mathbf{x}, \tau; \mathcal{K}^{(\text{iter})}, P_i)]$.
- 12: **end for**
- 13: Compute $\text{MI}(\mathcal{K}^{(\text{iter})})$ using the stored set $\{\mathcal{G}(\mathcal{K}^{(\text{iter})})\}$ and the Gauss-Hermite weights.
- 14: Update the design using the optimizer rule to increase MI:

$$\mathcal{K}^{(\text{iter}+1)} \leftarrow \text{UPDATE}(\mathcal{K}^{(\text{iter})}, \text{MI}(\mathcal{K}^{(\text{iter})})).$$

- 15: Check stopping criteria (e.g., $|\text{MI}(\mathcal{K}^{(\text{iter}+1)}) - \text{MI}(\mathcal{K}^{(\text{iter})})| < \epsilon$, or maximum iterations).
- 16: **end for**
- 17: Identify the design with maximum MI among all evaluated candidates:

$$\mathcal{K}^* = \arg \max_{\text{iter}=0, \dots, I} \text{MI}(\mathcal{K}^{(\text{iter})}).$$

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